Journal of Environmental Chemical Engineering 8 (2020) 103653

Contents lists available at ScienceDirect

Journal of Environmental Chemical Engineering

journal homepage: www.elsevier.com/locate/jece

One-pot synthesis of amorphous calcium phosphate/Fe_3O_4 composites and the application in the removal of cadmium

Trang Thuy Thuy Le Hoang[a], Fuangfa Unob[b], Soontorn Suvokhiaw[c], Nipaka Sukpirom[d,*]

[a] Department of Chemistry, Faculty of Science, Chulalongkorn University, Thailand
[b] Department of Chemistry, Faculty of Science, Chulalongkorn University, Thailand
[c] Department of Chemistry, Faculty of Science, Silpakorn University, Thailand
[d] Department of Chemistry, Faculty of Science, Chulalongkorn University, Bangkok, 10330, Thailand

ARTICLE INFO

Keywords:
Amorphous calcium phosphate
Hydroxyapatite
Cadmium removal
Citric acid
Magnetic composites

ABSTRACT

Amorphous calcium phosphate (ACP) is a promising material for cadmium removal due to its irregular surface structure. Nevertheless, the preparation of ACP has not been trivial owing to the facile transformation of the meta-stable state of ACP into the crystalline phase of hydroxyapatite (Hap). Herein we report a new preparation, by which ACP can be synthesized at ease through a one-pot synthesis. The utility of citric acid, together with, preformed Fe_3O_4 renders the formation of ACP/Fe_3O_4 magnetic composites, hampering the aggregation of calcium phosphates and hence hindering the transformation of ACP into undesired Hap. The well-controlled syntheses are illustrated and proved to be pH-dependent. The cadmium adsorption of the ACP/Fe_3O_4 composite is well-fitted with Langmuir isotherm giving a maximum capacity of 454.5 mg g^{-1} and shows a pseudo-second order kinetic behavior. The superior cadmium adsorption capacity of the composite is top ranked among previously reported adsorbents. The thorough investigation underpins the adsorption mechanism, whereby the specific replacement of Ca^{2+} ions in ACP network by Cd^{2+} is highly selective over other M(II) ions such as Co^{2+} and Ni^{2+}. Furthermore, the superparamagnetic nature of the composites allows a convenient separation of the materials after use.

1. Introduction

The contamination of toxic species in water sources has long been recognized as one of the most serious problems worldwide. The development of adsorbents to remove those species with effective cost, convenient process, high selectivity, high removal capacity and environmental benign is interesting due to the awareness on environmental issue [1]. The adsorbents could be categorized into low-cost pristine materials such as activated carbons [2], metal oxide nanoparticles [3], porous materials with high adsorption capacity such as zeolite [4], metal open-framework materials [5], surface-modified materials such as modified clays [6] and chitosan [7], composites with the combination of desired properties such as magnetic metal oxide nanoparticles and polymer nanocomposites [8], and conjugate nanomaterials exhibiting both selective adsorption and visual response [9,10]. Owing to the environmental benign properties and the high accessible surface areas of calcium phosphates (CPs), the materials have received plethora amount of attention in the field of water treatment and toxic removal among previously stated materials [11–18]. This work presents the composites with the combination of irregular surface of amorphous calcium phosphate network and a magnetic behavior of Fe_3O_4, aiming to obtain the adsorbents with high adsorption capacity, high selectivity and ease of separation.

Two types of CPs, including hydroxyapatite-based [11–17] and amorphous calcium phosphates [18] have been known for their potential applications in the removal of heavy metals, especially cadmium. It has been found that higher removal capacities of heavy metals were achieved from hydroxyapatite (Hap) with lower crystallinity and higher specific surface area [17]. Considering the correlation between crystallinity and removal capacity, amorphous calcium phosphate (ACP) could be a more promising sorbent than Hap.

ACP has various Ca/P ratios depending on the condition of the synthesis [19,20]. ACP precipitates normally contain a large amount of water including bound water (10–20 %) and water molecules in the space between the calcium phosphate clusters, called the Posner's clusters. The specific surface area of ACP was therefore surprising low

* Corresponding author.
E-mail addresses: trang.lehoang277@gmail.com (T.T.T.L. Hoang), fuangfa.u@chula.ac.th (F. Unob), s_suntornsc@hotmail.com (S. Suvokhiaw), nipaka.s@chula.ac.th (N. Sukpirom).

https://doi.org/10.1016/j.jece.2020.103653
Received 24 June 2019; Received in revised form 8 December 2019; Accepted 1 January 2020
Available online 03 January 2020
2213-3437/ © 2020 Published by Elsevier Ltd.

due to the obstruction of these water molecules [19]. Moreover, ACP is a meta-stable species, formed as a precursor of Hap in a basic condition. From a saturated calcium phosphate solution, ACP precipitates as an initial solid phase, then readily transforms to the stable crystalline materials, Hap [18,19,21–26]. Thus, the uses of special techniques, such as spray drying method [19], phosphorus-containing biomolecules as starting materials [18,25] or deep eutectic solvents [24] were necessary to prevent the transformation from ACP to Hap. Even though the potential of ACP as sorbents for toxic removal is realized, the use of ACP in this field has thus far been in their infancy majorly owing to their complicated preparations. Therefore, more facile syntheses of ACP are requisite to broaden the investigation and use of potent ACP in toxic removal avenues.

The use of CPs materials to remove heavy metals normally faces some difficulty in adsorbent separation [13,27,28]. A development of adsorbents that can be conveniently recovered is hence needed. The magnetic separation of materials from media is well known as an efficient, economic, and non-contaminated technology. Various magnetic composites used for metal ion adsorption including magnetic Hap were prepared and their easy separation using an external magnetic field was demonstrated [13,27,28]. However, the studies exploring the possibility of using magnetic ACP for heavy metals removal are scarce. ACP/$CoFe_2O_4$ composites were synthesized by co-precipitation method using block copolymer pluronic P123 as a template [29]. ACP/iron oxide composites were prepared using Na_2ATP as a stabilizer [30]. These magnetic ACP materials were used for biochemical applications. In this work, ACP/Fe_3O_4 composites were synthesized *via* a one-pot synthesis for cadmium removal. A one-pot synthesis has advantages including less time consuming and cost saving. Here we first report the preparation of ACP/Fe_3O_4 composites *via* a facile one-pot synthesis procedure.

Citric acid was applied in the one-pot synthesis of ACP/Fe_3O_4 composites. It is a weak organic tricarboxylic acid (pK_{a1} = 3.1, pK_{a2} = 4.8 and pK_{a3} = 6.4) [31], known as an inhibitor ceasing the transformation of ACP to Hap [22]. The acid and its anionic forms have a strong chelating ability to calcium ions depending strongly on the pH of the solution [32–34]. Moreover, citric acid could also stabilize the magnetite particles by using one carboxylate group to bind with magnetite surface, leaving other two carboxylates to prevent the aggregation or to provide the active sites for further surface modification [35]. The effect of factors related to the formation of ACP was studied and the mechanism of ACP formation was subsequently proposed. The cadmium removal behavior of these composites was also investigated.

2. Materials and methods

2.1. Materials

Ferrous sulfate heptahydrate ($FeSO_4$. $7H_2O$), Iron(III) chloride ($FeCl_3$), ammonia solution (30 %) (NH_4OH), calcium nitrate tetrahydrate ($Ca(NO_3)_2.4H_2O$), monoammonium phosphate ($NH_4H_2PO_4$), citric acid, cadmium sulfate 8/3-hydrate ($CdSO_4.8/3H_2O$), calcium carbonate ($CaCO_3$), HNO_3, NaOH, lead nitrate ($Pb(NO_3)_2$), cobalt(II) nitrate hexahydrate ($Co(NO_3)_2.6H_2O$), nickel nitrate hexahydrate ($Ni(NO_3)_2.6H_2O$), standard solution (1000 mg L^{-1}) of cadmium, lead, cobalt and nickel were analytical grade and used as received.

2.2. Synthesis of magnetite particles and calcium phosphates

Fe_3O_4 particles were synthesized based on the method of Suvokhiaw et al. [36] with some modifications. Under an inert atmosphere of nitrogen, 25 mL of 1.38 M Fe^{2+} (aq) and 25 mL of 2.76 M Fe^{3+} (aq) were mixed together to get the Fe^{2+}: Fe^{3+} molar ratio of 1:2. Concentrated ammonia solution was added to the mixed solution until the pH of solution reached 10.0. After stirring the mixture for 1 h, the black precipitates were separated using an external magnet and washed three times with deionized water. The as-synthesized Fe_3O_4 was dried at 60 °C under vacuum for 12 h.

The bulk calcium phosphates were synthesized by precipitation method. 20 mL of 2.10 M $Ca(NO_3)_2.4H_2O$ and 20 mL of 1.26 M $NH_4H_2PO_4$ (atomic Ca/P ratio of 1.67) were mixed together in aqueous solution. Then, concentrated ammonia solution was added until the pH of the suspension reached 10.5. After stirring the mixture for 1 h, the white precipitates were recovered by centrifugation and washed twice with deionized water. Finally, the received precipitates were dried at 90 °C for 12 h.

2.3. One-pot synthesis of CPs/Fe_3O_4 composites

The molar ratio of citric acid to Ca^{2+} and Fe^{3+} to Ca^{2+} were fixed at 0.25:1 and 0.22:1, respectively, throughout the experiments. Firstly, Fe_3O_4 particles were formed, using the same method as described in section 2.2 without collecting and washing steps. In the second step, 4 mL of 0.53 M citric acid was added into the preformed Fe_3O_4 mixture. At this step, the pH of the solution was adjusted from 10.0–2.3, 4.0, 6.0, 6.7, 7.9 or 9.5 using 2 M HNO_3. After continuous stirring for 20 min, 4 mL of 2.10 M Ca^{2+} and 4 mL of 1.26 M $H_2PO_4^-$ (Ca/P = 1.67) were added, as the third step. The pH of the suspension was adjusted to 10.5 using a concentrated ammonia solution. After stirring for 1 h, the obtained brown precipitates were collected using an external magnet and washed twice by deionized water and dried at 90 °C.

The effect of an initial Ca/P ratio was also studied. By maintaining the constant amount of $Ca(NO_3)_2.4H_2O$ and varying the amount of $NH_4H_2PO_4$, the composites with three ratios of Ca/P (1.5, 1.67 and 2) were synthesized using the above-mentioned method. The pH of the reaction mixture in the second step was fixed at 6.0.

2.4. Material characterization

The crystalline phases of the products were identified by XRD using a Rigaku, Dmax-2200 Ultima$^+$ diffractometer equipped with a monochromater and a Cu Kα radiation source. The surface area of the samples was measured by N_2 adsorption-desorption method using a BEL Japan BELSORP-mini 28SP instrument. The FTIR spectra were recorded as KBr pellets with a Thermo Scientific Nicolet 6700 FTIR spectrophotometer. Transmission electron microscopy (TEM) images were obtained using a Philips TECNAI 20 electron microscope. The surface morphology of the materials was monitored by SEM using a JSM-5800 LV scanning electron microscopy (Oxford Instrument Link ISIS series 300). A LECO CHNS-628 elemental analyzer was used to determine the percentage of carbon element in the final products in order to estimate the amount of citrate ions existing in the products. The magnetic hysteresis loops were recorded using an in-house developed VSM system [37], calibrated by nickel sphere from Lake Shore, model 730908. DSC results were recorded using a DSC1 Mettler Toledo. The measurements were performed by heating samples from 30 °C to 450 °C at a heating rate of 10 °C min^{-1} in air.

2.5. Adsorption experiments

The removal of Cd^{2+} by CPs/Fe_3O_4 composites was evaluated using batch experiments. The bottles containing 10 mg of sorbents and 10.0 mL of Cd^{2+} each were shaken at 30 rpm. The sample solutions were collected by separating the solid through 0.45 μm syringe filters.

The effect of solution pH on cadmium adsorption (Cd^{2+} initial concentration of 250 mg L^{-1}) was studied in the initial pH range of 2–7. The pH of the initial solution of Cd^{2+} was adjusted to desire values using 0.01 M HNO_3 or 0.01 M NaOH.

To study the effect of ionic strength, 0.003-0.5 mol L^{-1} of $NaNO_3$ was added to Cd^{2+} solutions (350 mg L^{-1}). The pH of the initial solution of Cd^{2+} was adjusted to 5.0. The suspensions were shaken at 25 °C for 15 h.

To investigate the selectivity of the composites, the binary metal solutions of Cd^{2+} with Co^{2+}, Ni^{2+} and Pb^{2+} were prepared. The initial pH of all metals solutions was 5.0. The initial concentrations of each metal ions were kept approximately at 3 M. The suspensions were shaken at 25 °C for 15 h.

The experiments of sorption isotherms investigated under Cd^{2+} initial concentration in a range from 0 – 778 mg L^{-1} shaken for 24 h and sorption kinetics evaluated at various sorption times (5 – 420 min) were performed.

The initial and equilibrium concentration of Cd^{2+} was determined by an inductively coupled plasma atomic emission spectrometer (ICP-OES) model iCAP 6500 series (Thermo Fisher Scientific). The capacity of the material in cadmium adsorption (q_t (mg g^{-1})) was calculated by Eq. 1;

$$q_t = [(C_0 - C_t)V]/m \quad (1)$$

where C_0 and C_t are the initial concentration and the concentration at given time t (mg L^{-1}) of Cd^{2+}, respectively. V is the volume of solution (L), and m is the weight of sorbents (g).

3. Results and discussion

3.1. Preparation and characterization of the materials

To prepare CPs/Fe_3O_4 composites, the one-pot synthesis was conducted in three consecutive steps: (1) the formation of Fe_3O_4 nanoparticles (MNPs), (2) the complexation of citrate ions on MNPs surface, and (3) the precipitation of CPs/Fe_3O_4 composites. The effect of various parameters on the formation of CPs/Fe_3O_4 composites was investigated.

3.1.1. Effect of pH of the solution

In previous studies, citric acid has been utilized in the modification of the magnetite surface as well as the formation of complexes with calcium phosphates [22,32–35]. In this work, citric acid was added into the mixture during step (2) of the one-pot synthesis and the pH of the mixture solution was varied to observe its effect on the formation of calcium phosphates. The XRD patterns of CPs/Fe_3O_4 composites derived at various pH are shown in Fig. 1. The characteristic peaks of Fe_3O_4 (PDF 86–1361) appear in all patterns. The detailed investigation using XRD provided the evidence of Hap in composites prepared at the pH higher than the pK_{a3} of citric acid (6.7–9.5). In a stark contrast, no Hap was marked in any composite derived at the pH lower than the pK_{a3} of citric acid (2.3–6.0). This observation could imply the successful synthesis of ACPs/Fe_3O_4 under the optimal acidic conditions.

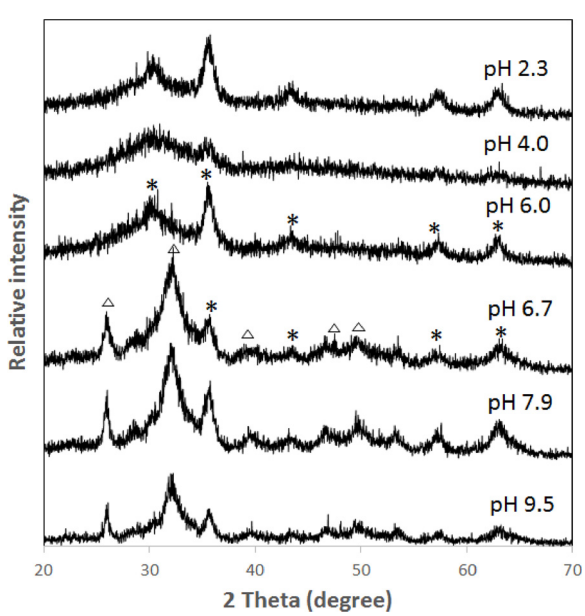

Fig. 1. XRD patterns of CPs/Fe_3O_4 synthesized at various pH of the reaction mixture in the step (2) (△: Hap, *: Fe_3O_4).

TEM and SEM images of the bare Fe_3O_4 (Fig. 2 (a) and (b)) show the agglomeration of nano-sized particles. For the composites, ACP/Fe_3O_4 (pH 6.0) and Hap/Fe_3O_4 (pH 9.5) were chosen as the representatives of the composites containing ACP and Hap, respectively. The TEM image of ACP/Fe_3O_4 (Fig. 2(c)) suggests the particle aggregation. Fe_3O_4 particles were covered and linked by the irregular shape of ACP phase, resulting to the mossy surface as seen in the SEM image (Fig. 2(d)). On the other hand, the needle-shaped particles of Hap and the nano-sized Fe_3O_4 aggregate together in Hap/Fe_3O_4 (Fig. 2(e) and 2(f)).

To determine the content of calcium phosphates and magnetite in the materials, samples were digested in acid and the concentrations of calcium, phosphorus, and iron were measured by ICP-OES. The content of citrate represented by carbon content was observed using elemental analysis technique. The elemental content results are shown in Table 1. According to the weight percentages of iron detected, the content of MNPs was in a range from 9.8–19.9 wt% of composites depending on the pH during the synthesis. The average amount of MNPs in the ACP/Fe_3O_4 composites was higher than those in Hap/Fe_3O_4. According to the percentages of carbon detected, 11.9 wt% - 16.7 wt% citrate ions were retained in the prepared composites. The presence of citrate ions in the final products were confirmed by FTIR (Fig. S1). The bands at 1633 cm^{-1} and 1430 cm^{-1} could be assigned as $n_{asym}(COO^-)$ and $n_{sym}(COO^-)$ of citrate ions, respectively [31,34]. These citrate ions in the composites would help maintaining the structure of the materials as discussed in the synthesis mechanism. The weight percentages of calcium and phosphorus indicating the amount phosphate were found to be in a range of 18.1–25.0 wt% and 8.9–11.8 wt%, respectively. The calcium and phosphorus contents slightly increased when the pH in the step (2) increased from 2.3–9.5. This is in line with the previous observation, whereby a higher pH condition could induce the formation of calcium phosphates better than that obtained at lower pH [38].

The specific surface area and the capacity of the materials in cadmium adsorption are shown in Table 1. Among all the parameters presented in this table, the capacity of the materials in cadmium adsorption depended strongly on the phase of calcium phosphates. ACP/Fe_3O_4 composites possessed higher adsorption capacities than Hap/Fe_3O_4 composites although they had lower surface area. The large difference in adsorption capacities between ACP composites and Hap composites pointed to the effect of crystallinity on the adsorption efficiency. In the amorphous phase (ACP), Ca^{2+} would be substituted by Cd^{2+} easier than those in a crystalline phase (Hap) due to the presence of defects and dangling bonds normally found in amorphous materials [39,40]. The mechanism of cadmium adsorption will be discussed hereafter. The unanticipated correlation between the surface area and cadmium removal ability could be explained by the different interaction of the water-containing ACP-based materials on the N_2 adsorption and on the Cd^{2+} adsorption. For the determination of the specific surface area of ACP/Fe_3O_4, the physical adsorption occurred between the solid and nitrogen gas molecules. Therefore, the tightly bound interspace water in the ACP/Fe_3O_4 materials would obstruct the adsorption of nitrogen molecules and thus reduce the measured specific surface area (Fig. S2) [19]. In the cadmium adsorption process, sorbents were suspended into the Cd^{2+} solution. The metal ions would diffuse inside the pore easily through the water inside the structure of ACP-based materials that acted as the medium for the mass transfer in the sorption process. The cadmium adsorption capacities of ACP/Fe_3O_4 composites synthesized at pH 6.0 and pH 4.0 were not significantly different, and they were higher than that of the composite synthesized at pH 2.3. Therefore, pH of 6.0 was chosen for the preparation of the materials for further investigation.

3.1.2. Effect of Ca/P ratio

As previously reported, ACP existed in the wide range of Ca/P ratios [19]. To synthesize the magnetic ACP in this study, the Ca/P ratio was

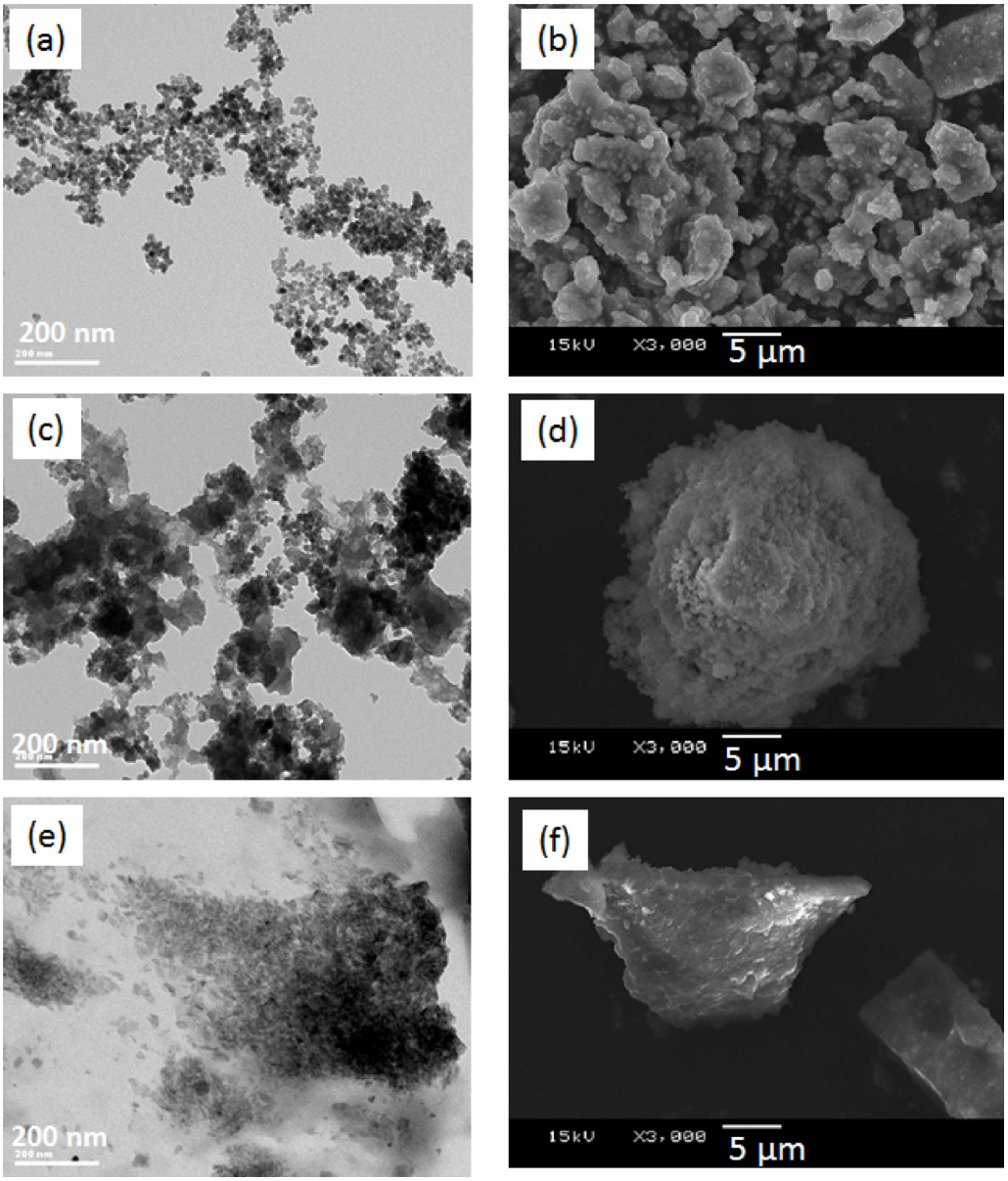

Fig. 2. TEM (a) and SEM (b) images of bare Fe_3O_4, TEM (c) and SEM (d) images of ACP/Fe_3O_4 (synthesized at pH 6.0) and TEM (e) and SEM (f) images of Hap/Fe_3O_4 (synthesized at pH 9.5).

Table 1
Surface area, elemental content and capacity in cadmium adsorption of the CPs/Fe_3O_4 composites synthesized at various pH in the step (2).

pH	Phases of CPs	S_{BET}[a] (m^2 g^{-1})	C[b] (wt%)	Ca[c] (wt%)	P[c] (wt%)	Fe[c] (wt%)	Ca/P[c]	Cadmium adsorption capacity[d] (mg g^{-1})
pH 2.3	ACP	119.0	5.63	18.1	8.90	17.8	1.57	275.1 ± 10.2
pH 4.0	ACP	81.9	6.36	19.1	9.31	11.8	1.56	299.0 ± 13.7
pH 6.0	ACP	88.2	5.55	20.4	10.1	12.7	1.56	292.9 ± 2.6
pH 6.7	Hap	156.5	5.31	22.3	10.9	8.79	1.58	154.8 ± 7.8
pH 7.9	Hap	192.9	4.96	23.5	11.1	13.3	1.63	136.3 ± 2.1
pH 9.5	Hap	208.3	4.52	25.0	11.8	11.4	1.63	138.7 ± 0.9

[a] from N_2 isotherm using the BET model.
[b] from CHN elemental analysis.
[c] Samples were digested in acid and the concentration of calcium, phosphorus, and iron were measured by ICP-OES.
[d] The adsorption condition: 10 mg of sorbents, 10 mL of 390 mg L^{-1} Cd^{2+}, pH of 5.0. The concentration of Cd^{2+} was measured by ICP-OES.

varied in the range of 1.5 – 2.0. The XRD patterns of materials with different Ca/P ratios are illustrated in Fig. 3. Only Fe_3O_4 phase appeared in all XRD patterns, indicating that the calcium phosphate of all three Ca/P ratios was an amorphous phase. The irregular shape of ACP particles linking together and becoming random network that covered the Fe_3O_4 particles could be clearly perceived in the TEM image of ACP/Fe_3O_4 (Fig. S3). Furthermore, all three ACP/Fe_3O_4 composites possessed closed values of specific surface areas and cadmium

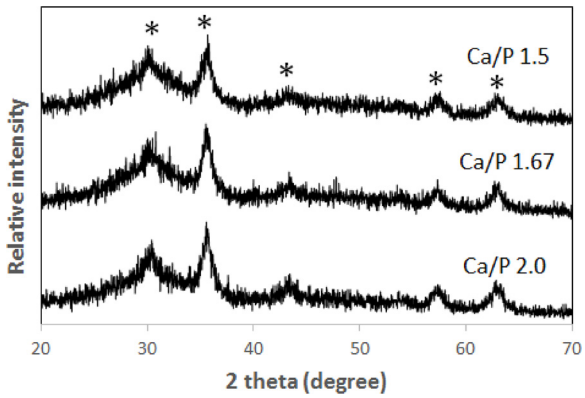

Fig. 3. XRD patterns of ACP/Fe_3O_4 synthesized with different Ca/P ratio (*: Fe_3O_4).

adsorption capacities, as summarized in Table 2. Therefore, the Ca/P ratios in this range show negligible effect either on the phase formation or the cadmium adsorption.

3.1.3. Magnetic properties of the materials

The magnetic properties of Hap/Fe_3O_4 and ACP/Fe_3O_4 were compared with that of Fe_3O_4 (Fig. 4). All hysteresis curves exhibited nearly zero values of magnetic remanence and coercivity, showing their superparamagnetic property [28,41]. The saturation magnetization parameter (M_s) of Fe_3O_4 was 61.4 emu g^{-1}, while those of Hap/Fe_3O_4 (consisting of 15.8 wt% Fe_3O_4) and ACP/Fe_3O_4 (consisting of 15.9 wt% Fe_3O_4) were 8.94 emu g^{-1} and 7.80 emu g^{-1}, respectively. The non-magnetic phase of ACP or Hap may cause the lower saturation magnetization of the composites owing to the feasible separation of each magnetite particles. The observation replicates the phenomenon, of which under an external magnetic field the magnetic induction among different magnetic domains was interrupted by a non-magnetic phase [36,42,43].

3.1.4. Mechanism of the synthesis process

The schematic illustration of the formation of ACP/Fe_3O_4 and Hap/Fe_3O_4 is shown in Fig. 5. There were two types of the formation of magnetic calcium phosphates (*i.e.* Hap and ACP) depending on the pH of the reaction mixture when citric acid was added to the preformed magnetic particles (step (2)).

In the pH range lower than the pK_{a3} of citric acid (less than 6.4), Fe_3O_4 particle had positive surface charge [44,45]. There would be the electrostatic interaction between the negative charge of the partly-deprotonated species of citric acid ($H_2C_6H_5O_7^-$ and $HC_6H_5O_7^{2-}$) and the positive charge on Fe_3O_4 surface. Therefore, the anions were coated on the surface of Fe_3O_4 and acted as steric stabilizer of magnetite particles. After that, when Ca^{2+} and PO_4^3 were added, another end of the citrate anions would bind with Ca^{2+}, linking Ca^{2+} to the surface of magnetite particles. As a result, the Ca^{2+} ions were attached on the surface of magnetite, and the meta-stable phase of amorphous calcium phosphate was formed and maintained on the particle surface. On the other hand, the synthesis of calcium phosphates with only magnetite or citric acid under similar conditions always yielded the Hap phase. The cooperation of magnetite and citrate species was therefore needed for stabilizing ACP.

In the range of pH higher than the pK_{a3} of citric acid (6.7–9.5), the charge on the surface of magnetite particles became negative or less positive [44,45]. Therefore, the interaction between fully-deprotonated citrate ion ($C_6H_5O_7^{3-}$) and magnetite surface became weaker. While citrate ions bound and chelated strongly with Ca^{2+} ions, and the crystalline phase of Hap particles would form and further deposit or aggregate with magnetite particles. The hypothesis of stronger chelation between citrate and Ca^{2+} at higher pH was supported with the reported binding constants of citrate ion and Ca^{2+} ($K = 7.4 \times 10^3$ at the pH greater than 6.4, and K = 832 or K = 28.2 at the pH less than 6.4) [32].

In order to understand how citrate ions bonded in the materials, the thermal stability of the materials synthesized with and without citric acid were evaluated and compared by DSC technique (Fig. 6). Hap and Hap/Fe_3O_4 samples synthesized without citric acid were thermally stable up to 450 °C as no peak was observed in the DSC results. On the other hand, for the analysis of the other samples containing citrate ions, the endothermic peaks appeared in range of 143 °C – 167 °C corresponding to the melting temperature of citrate ions [46]. The shift of melting temperature of citrate in DSC results could predict the bonding strength of citrate ion with its surroundings. The degree of shifting to lower melting point was as follows: ACP/Fe_3O_4 (with citric acid) > Hap/Fe_3O_4 (with citric acid) > Hap (with citric acid). This might be explained that citrate ions could bind with crystalline phase of Hap in more ordered fashion, as compared with the composites. With the same reason, the random arrangement of amorphous phase would shift the melting point of citrate ions to the lowest temperature.

3.2. Sorption study

As mentioned in Section 3.1 (Table 1), the high cadmium adsorption capacity relies mainly on the amorphous phase of calcium phosphate; therefore, ACP/Fe_3O_4 (Ca/P = 1.5, obtained at pH of 6.0) was selected as the sorbent for sorption study. The adsorption properties of the sorbent including adsorption kinetics and adsorption behavior at equilibrium were studied. The adsorption mechanism was also investigated.

3.2.1. The effect of pH

The effect of solution pH on cadmium adsorption was studied in the initial pH range of 2–7. As shown in Fig. 7 from pH 3–7, the cadmium adsorption capacity remained constant, indicating that the solution pH has no effect on the adsorption process. The dramatic drop of the capacity at the pH lower than 3 was due to the dissolution of ACP in strong acidic solution.

It should be noted that starting with the initial solution pH of 3–7, the pH at equilibrium of these solutions was *ca.*6 (5.69–6.15) despite the different initial pH values owning to the basic nature of calcium phosphates. At lower pH, sparingly soluble calcium phosphates become slightly soluble, increasing the pH of solution as a result [47,48]. It is

Table 2
Surface area, the elemental content, and the capacity in cadmium adsorption of the ACP/Fe_3O_4 composites synthesized with different Ca/P ratio.

Samples	Phase of CPs	S_{BET}[a] ($m^2 g^{-1}$)	C[b] (wt%)	Ca[c] (wt%)	P[c] (wt%)	Fe[c] (wt%)	Ca/P[c]	Cadmium adsorption capacity[d] (mg g^{-1})
Ca/P = 1.5	ACP	88.4	6.16	19.9	10.6	11.5	1.45	304.5 ± 7.4
Ca/P = 1.67	ACP	88.2	6.52	20.4	10.1	12.7	1.56	292.9 ± 2.6
Ca/P = 2.0	ACP	77.0	8.21	19.3	8.62	15.0	1.73	302.6 ± 3.4

[a] from N_2 isotherm using the BET model.
[b] from CHN elemental analysis.
[c] Samples were digested and the concentration of calcium, phosphorus and iron were measured by ICP-OES.
[d] The adsorption condition: 10 mg of sorbents, 10 mL of 390 mg L^{-1} Cd^{2+}, pH of 5.0. The concentration of Cd^{2+} was measured by ICP-OES.

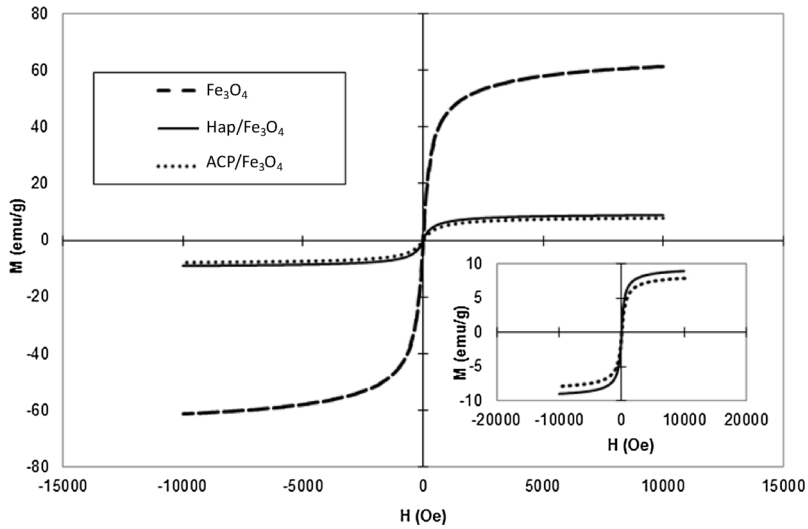

Fig. 4. Magnetization hysteresis curves of Fe_3O_4, Hap/Fe_3O_4 (pH 9.5, Ca/P = 1.67) and ACP/Fe_3O_4 (pH 6.0, Ca/P = 1.5) (The inset shows the low field region of the magnetization curve of Hap/Fe_3O_4 and ACP/Fe_3O_4.).

likely that at this pH range, the active sites for Cd^{2+} adsorption on ACP exhibited the same nature and hence the uptake of Cd^{2+} was not affected. The initial pH of 5 was adopted in further adsorption experiments.

3.2.2. Adsorption kinetics

The adsorption of Cd^{2+} by the material was observed at different contact time varied from 5 min to 7 h (Fig. 8a). The adsorption occurred rapidly in the first 40 min and reached the equilibrium after 2 h. In order to determine the rate of the adsorption, the pseudo-first order model (Eq.2) and pseudo-second order model (Eq.3) [27] were applied to the experimental kinetics data. The adsorption capacities (mg g^{-1}) observed at contact time t, and at equilibrium are presented by the variables q_t and q_e, respectively. The pseudo-first order rate constant (k_1) and pseudo-second order rate constant (k_2) were determined by linear plotting of the experimental data. It was observed that the kinetics data were fitted with the pseudo-second order model ($R^2 = 0.999$) (Fig. 8b) better than the pseudo-first order model ($R^2 = 0.925$) (Fig. S4). The pseudo-second order rate constant (k_2) was 3.6×10^{-4} g mg^{-1} min^{-1}. The adsorption capacity at equilibrium calculated using pseudo-second order model ($q_{e,cal}$) was 312.5 mg g^{-1}, which was close to the experimental value ($q_{e,exp}$) of 294.1 mg g^{-1}, indicating a good fit of the model to the experimental data. Using the k_2 value, the initial rate ($h = k_2 q_e^2$) of cadmium adsorption on ACP/Fe_3O_4 was calculated to be 0.112 mg g^{-1} min^{-1}.

$$\log(q_e - q_t) = \log q_e - \frac{k_1}{2.303}t \quad (2)$$

$$\frac{t}{q_t} = \frac{1}{k_2 q_e^2} + \frac{t}{q_e} \quad (3)$$

3.2.3. The effect of ionic strength

To understand the adsorption nature of the sorbent, the effect of ionic strength on the adsorption of cadmium was studied using $NaNO_3$ solution. The results in Fig. 9 indicate that there was no statistically significant difference in removal efficiencies when the solution contained $NaNO_3$ in the range of 0 – 0.5 M (one way ANOVA, P = 0.07) at 95 % confident interval (α = 0.05.). The result could imply that the main factor of adsorption mechanism was not a simply coulombic interaction [49].

3.2.4. Selectivity

The presence of competitive cations such as Co^{2+}, Ni^{2+}, and Pb^{2+} could affect the adsorption performance of ACP/Fe_3O_4 for the selective removal of Cd^{2+} ions. Fig. 10 shows the removal percentage of Cd^{2+} from a solution containing single metal ions and those containing binary metal (2+) ions. In the case of Cd^{2+}-Co^{2+} and Cd^{2+}-Ni^{2+} solutions, ACP/Fe_3O_4 could adsorb a certain amount of other cations; however, the ratios of removal percentages were 0.25 and 0.05 for Co^{2+} and Ni^{2+} respectively, when compared with that of Cd^{2+}. The

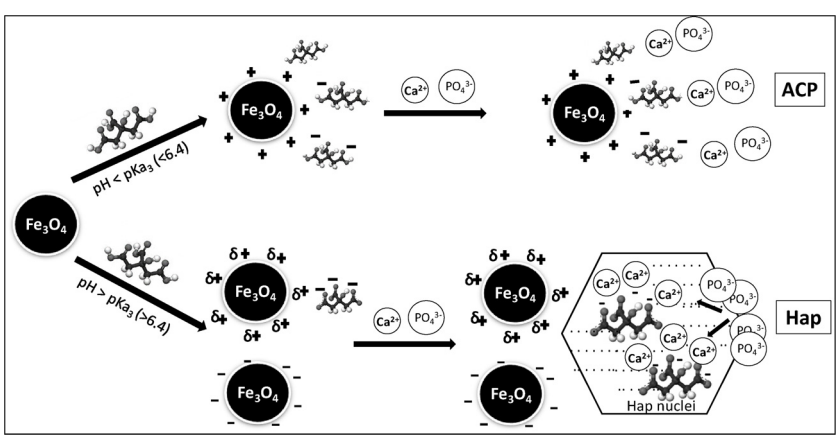

Fig. 5. Schematic illustration of the formation of ACP/Fe_3O_4 and Hap/Fe_3O_4 composites.

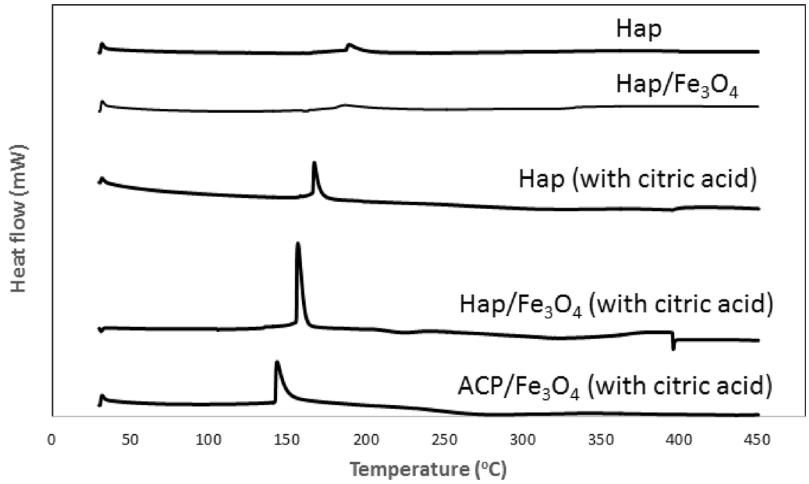

Fig. 6. DSC curves of Hap, Hap/Fe_3O_4 synthesized without citric acid, and the materials synthesized in the presence of citric acid including Hap, Hap/Fe_3O_4 (pH 6.0, Ca/P = 1.67), and ACP/ Fe_3O_4 (pH 6.0, Ca/P = 1.5).

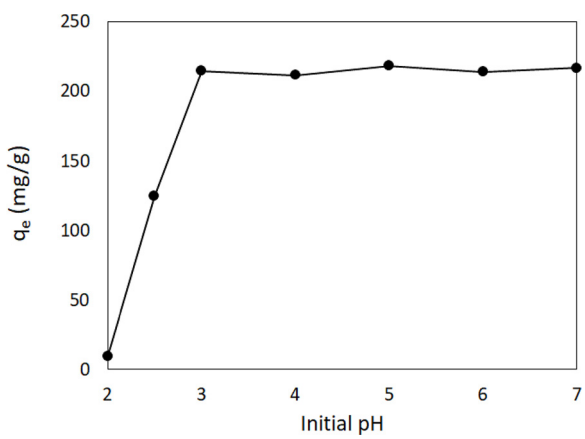

Fig. 7. Effect of initial solution pH on cadmium removal ability of ACP/Fe_3O_4. (250 mg L^{-1} Cd^{2+})

reason is due to the size compatibility of Cd^{2+} at the adsorption site. The ionic radius of Cd^{2+} (95 pm) is comparable to that of Ca^{2+} (100 pm); however, is much larger than that of Co^{2+} (74 pm) and Ni^{2+} (69 pm). The solution of Cd^{2+}-Pb^{2+}, on the other hand, exhibited the significantly lower adsorption of Cd^{2+} by 50 %, with the disappearance of Pb^{2+} in the solution. The explanation could be that lead ions disrupted the calcium phosphate network and precipitated as $Pb_{10}(PO_4)_6(OH)_2$, as revealed by the XRD pattern of used ACP/Fe_3O_4 (Fig. S5). These results along with the results from the effect of solution ionic strength indicate that ACP/Fe_3O_4 exhibited high selectivity towards Cd^{2+} adsorption attributed to the size of the adsorption site in the material.

3.2.5. Adsorption isotherms

To investigate the adsorption behavior at equilibrium, the adsorption isotherms experiments were performed. The effect of cadmium concentration on the capacity of ACP/Fe_3O_4 in cadmium adsorption is demonstrated in Fig. 11a. The cadmium adsorption capacity increased with the increasing cadmium initial concentration. The Langmuir (Eq.4) and Freundlich (Eq.5) isotherm models [14,27] were applied to analyze the adsorption behavior of this material. The Langmuir constant related to energy of adsorption (b), the adsorbent maximum adsorption capacity (q_m), the Freundlich constant (n) and a constant related to adsorption capacity (K_f) could be calculated from the linear equation obtained from experimental data plots. According to the results, the Langmuir model fitted with the obtained adsorption data

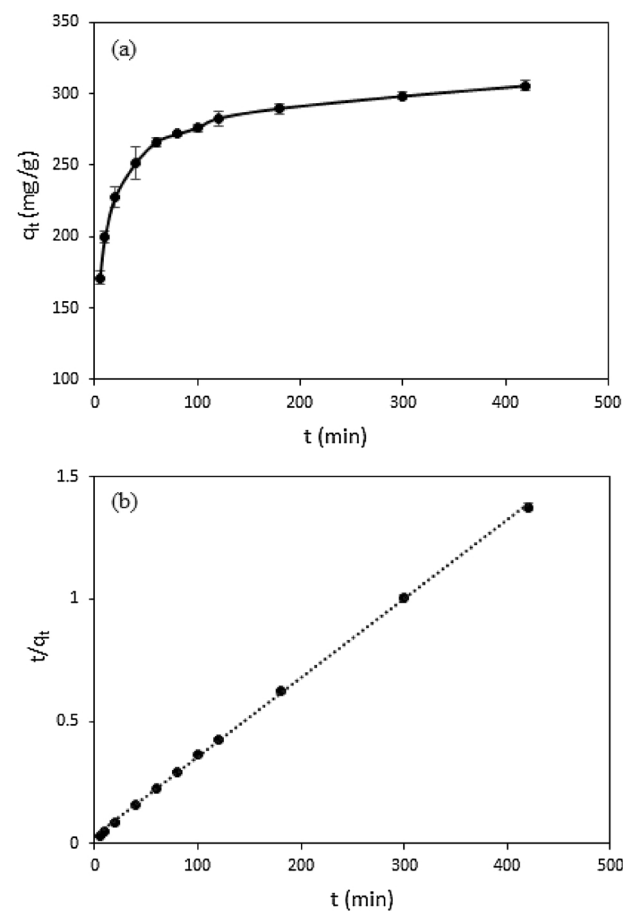

Fig. 8. Cadmium adsorption kinetics on ACP/Fe_3O_4 composite (Ca/P = 1.5) (a) and pseudo-second order model linear plot (b).

(R^2 = 0.993, Fig. 11b) better than the Freundlich model (R^2 = 0.978, Fig. S6). The Langmuir constant was 0.04 L mg^{-1} and the maximum adsorption capacity was 454.5 mg g^{-1}.

$$\frac{C_e}{q} = \frac{C_e}{q_m} + \frac{1}{bq_m} \tag{4}$$

$$\log q = \log K_f + \frac{1}{n}\log C_e \tag{5}$$

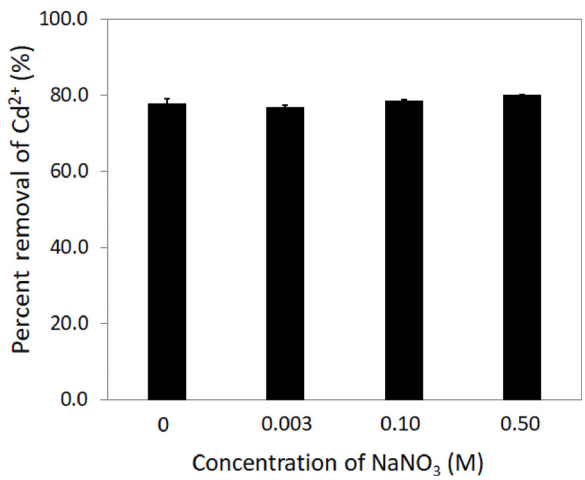

Fig. 9. Effect of ionic strength on cadmium removal ability of ACP/Fe_3O_4 (350 mg L^{-1} Cd^{2+}).

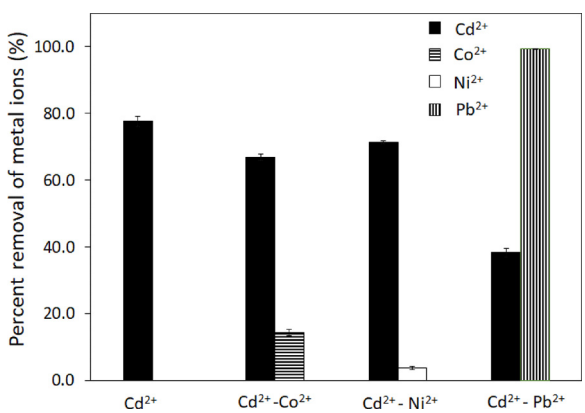

Fig. 10. Selective adsorptions towards Cd^{2+} in the aqueous solution containing only Cd^{2+} (3.2 M), and in the solution containing binary metal ions, including Cd^{2+}-Co^{2+} (3.0 M Cd^{2+} and 3.2 M Co^{2+}), Cd^{2+}-Ni^{2+} (3.2 M Cd^{2+} and 3.4 M Ni^{2+}), and Cd^{2+}-Pb^{2+} (2.9 M Cd^{2+} and 2.6 M Pb^{2+}).

3.2.6. Adsorption mechanism

By obtaining a good fitting of the Langmuir model to the adsorption isotherms, it confirms the monolayer coverage of Cd^{2+} ions onto ACP/Fe_3O_4 [50]. The reaction of the sorbent and sorbate only took place on the active sites (Ca^{2+}) on the surface of material. In order to confirm the adsorption mechanism, the release of Ca^{2+} from materials into the solutions during the adsorption of Cd^{2+} was monitored. The relationship between the amount of Ca^{2+} released and the amount of Cd^{2+} adsorbed was investigated (Fig. S7). The results indicated that the amount of Cd^{2+}adsorbed on the materials was directly proportional to that of Ca^{2+} released from the materials. The ratio of the amount of Cd^{2+} adsorbed (mmol g^{-1}) to the amount of Ca^{2+} released (mmol g^{-1}) was found in a range from 0.77 to 1.15 which was close to unity. Furthermore, the structure of ACP/Fe_3O_4 before and after the adsorption process was examined by XRD (Fig. S5). The XRD pattern of the material after being used in cadmium adsorption was similar to that of the pristine materials. This result indicated that the structure of magnetic amorphous calcium phosphate remained intact through the sorption process. With the unchanged structure and the direct ration of Cd^{2+} and Ca^{2+} in agreement with other studies [14,51], this work confirmed the ion exchange mechanism between Ca^{2+} in the ACP and Cd^{2+} in the solution as shown in equation (6).

$$[\text{ACP-Ca}^{2+} + \text{Cd}^{2+} \rightarrow \text{ACP-Cd}^{2+} + \text{Ca}^{2+}] \quad (6)$$

Based on this mechanism, the solution ionic strength would have no influence on the adsorption, since the Cd^{2+} adsorption took place by networking with phosphate rather than by coulombic interaction at the outer surface. In addition, the adsorption selectivity is achieved as Cd^{2+} possesses the matching size to Ca^{2+}; therefore, the removal of Cd^{2+} is more favorable compared to the smaller cations such as Co^{2+} and Ni^{2+}.

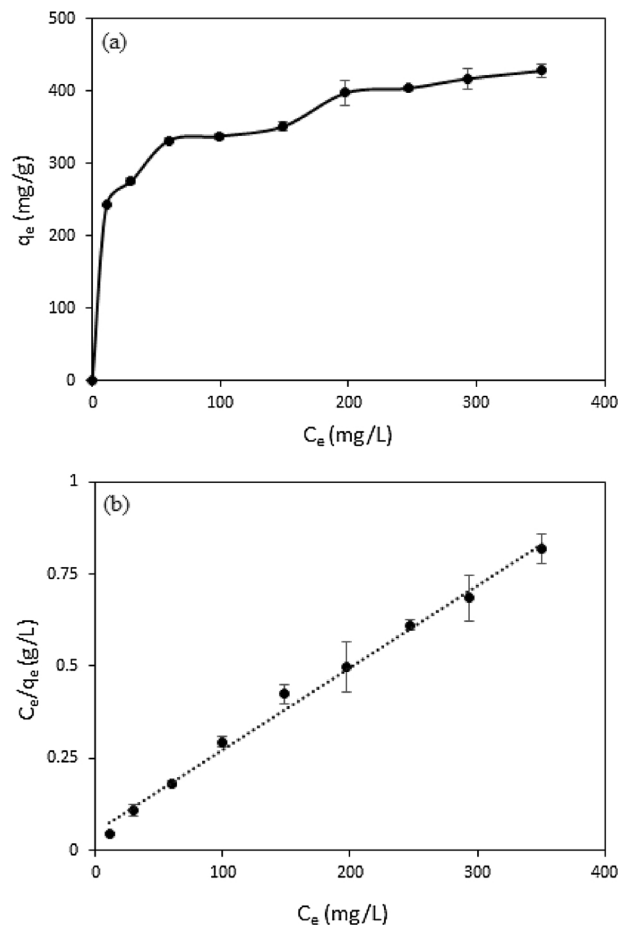

Fig. 11. Isotherm of cadmium adsorption on ACP/Fe_3O_4 composite (Ca/P = 1.5) (a) and Langmuir isotherm model linear plot (b).

The capacity of the ACP/Fe_3O_4 in Cd^{2+} ions adsorption is compared with precedently reported materials (Table 3). Among all of these materials, the ACP/Fe_3O_4 composite obtained by the proposed one-pot synthesis method shows intrinsically high adsorption capacity. When compared to crystalline Hap, the ACP provided higher adsorption capacity. In amorphous materials, there were defects and dangling bonds [39,40,52,53] and Ca^{2+} ions might be released and replaced by Cd^{2+} ions easier than those in a crystalline phase. Therefore, the materials having lower crystallinity would exhibit higher Cd^{2+} adsorption capacity. This evidence makes ACP an interesting material in respect to its ion exchange ability. Moreover, in the presence of the magnetic particles, the ACP/Fe_3O_4 composites in this work have also facilitated the convenient adsorbent separation.

3.2.7. Desorption and reuse

To investigate the reuse of ACP/Fe_3O_4, the desorption of Cd^{2+} from the used material was performed using either 0.05 M $Ca(NO_3)_2$ or 0.05 M $Ca(NO_3)_2$ in 0.09 M thiourea (Fig. 12). The desorption of Cd^{2+} ions by replacing with the excess of Ca^{2+} ions alone and with the co-operation of thiourea resulted to the desorption percentage of 13.6 and 22.0 respectively. Due to the adsorption mechanism of Cd^{2+} on ACP/Fe_3O_4 that went through the replacement of Ca^{2+} on the network of calcium phosphate, the complete desorption of Cd^{2+} ions and hence the reusibility of the sorbent at its full-potential would be unattainable. The recovered adsorbent from the $Ca(NO_3)_2$/thiourea solution showed a

Table 3
The comparison of cadmium adsorption capacities of this work with other works.

Materials	Method for the preparation	Cadmium adsorption capacities (mg g^{-1})	References
Hydroxyapatite	Precipitation	67.6*	[51]
Hydroxyapatite	Hydrothermal technique	260.42	[14]
Hydroxyapatite/Fe_3O_4	Precipitation	220.8*	[13]
Amorphous calcium phosphate	Microwave-assisted hydrothermal method	366	[18]
4-tert-Octyl-4-((phenyl) diazenyl)phenol anchored with mesoporous silica	Sol-gel synthesis and immobilization of the ligand	148.32	[54]
Sugarcane bagasse modified with Triethylenetetramine	Reflux and filtration	313	[55]
Broad bean peel	Abundantly available	147.71	[56]
Amorphous calcium phosphate/Fe_3O_4	One-pot synthesis	454.5	This work

* calculated from the published paper.

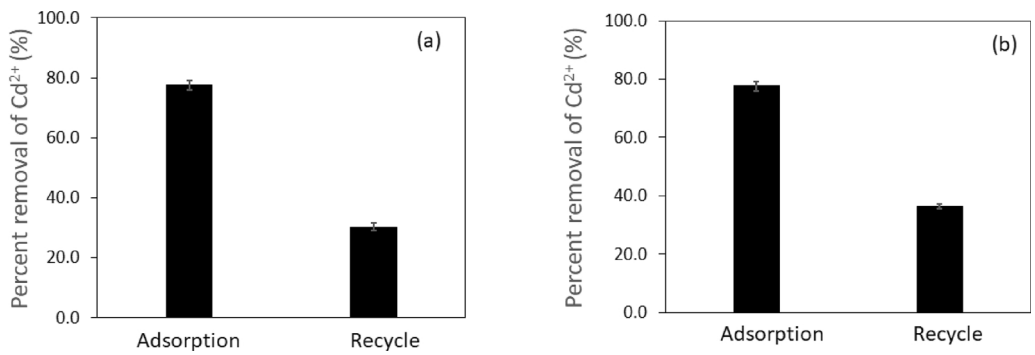

Fig. 12. Adsorption, desorption, and recycle of Cd^{2+} on ACP/Fe_3O_4, the desorption process performed by stirring 20 mg of Cd-adsorbed ACP/Fe_3O_4 in 20 mL of 0.05 M $Ca(NO_3)_2$ (a) and stirring 20 mg of Cd-adsorbed ACP/Fe_3O_4 in 0.05 M $Ca(NO_3)_2$ and 0.09 M thiourea (b).

Cd^{2+} removal percentage of 36.4 %, compared to 77.6 % performed by the freshly prepared ACP/Fe_3O_4. However, the recovered sorbents could adsorb the amount of Ca^{2+} exceeding the desorption capacity. This is possible due to the availability of the remaining Ca^{2+} in the calcium phosphate sorbent for the replacement with Cd^{2+}.

4. Conclusion

The amorphous calcium phosphate/Fe_3O_4 composite was successfully prepared *via* a simple precipitation method. Fe_3O_4 could prevent the aggregation of calcium phosphate nuclei, and citric acid was found to inhibit the transformation of amorphous calcium phosphate into hydroxyapatite. The degree of transformation of ACP to Hap was totally pH-dependent and directly involved the forms of citric acid during the synthesis, namely the fully-deprotonated citrate ions promoted the formation of Hap, while the partly deprotonated species encouraged the production of ACP.

Higher cadmium adsorption capacity was achieved from the magnetic composites of ACP. The isotherms and kinetics of Cd^{2+} adsorption were fitted with Langmuir model and pseudo-second order model, respectively. The maximum capacity achieved from the Langmuir model was 454.5 mg g^{-1}. This composite is a promising material having many advantages, such as having simple preparation, low cost, high Cd^{2+} removal capacity, high selectivity and convenient separation.

Author contributions

Trang Thuy Thuy Le Hoang: Study conception and design; Acquisition of data; Analysis and interpretation of data; Drafting of manuscript; Critical revision

Fuangfa Unob: Study conception and design; Analysis and interpretation of data; Drafting of manuscript; Critical revision

Soontorn Suvokhiaw: Study conception and design; Acquisition of data; Analysis and interpretation of data; Drafting of manuscript

Nipaka Sukpirom: Study conception and design; Acquisition of data; Analysis and interpretation of data; Drafting of manuscript; Critical revision

Declaration of Competing Interest

The authors declare that there is no conflict of interest.

Acknowledgement

The work was supported by the scholarships from the Graduate School, Chulalongkorn University to commemorate 72nd Anniversary of his Majesty King BhumibalaAduladeja and the 90th Anniversary Chulalongkorn University Fund (Ratchadaphiseksomphot Endowment Fund). We also thank Junjuda Unruangsri for making some constructive suggestions in our work.

Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:https://doi.org/10.1016/j.jece.2020.103653.

References

[1] N.B. Singh, G. Nagpal, S. Agrawal, Rachna, Water purification by using adsorbents: a review, Environ. Technol. Inno. 11 (2018) 187–240, https://doi.org/10.1016/j.eti.2018.05.006.
[2] G.Z. Kyzas, G. Bomis, R.I. Kosheleva, E.K. Efthimiadou, E.P. Favvas, M. Kostoglou, A.C. Mitropoulos, Nanobubbles effect on heavy metal ions adsorption by activated carbon, Chem. Eng. J. 356 (2019) 91–97, https://doi.org/10.1016/j.cej.2018.09.019.
[3] L.P. Lingamdinne, Y.Y. Chang, J.K. Yang, J. Singh, E.H. Choi, M. Shiratani, J.R. Koduru, P. Attri, Biogenic reductive preparation of magnetic inverse spinel iron oxide nanoparticles for the adsorption removal of heavy metals, Chem. Eng. J. 307 (2017) 74–84, https://doi.org/10.1016/j.cej.2016.08.067.
[4] M. Hong, L. Yu, Y. Wang, J. Zhang, Z. Chen, L. Dong, Q. Zan, R. Li, Heavy metal adsorption with zeolites: the role of hierarchical pore architecture, Chem. Eng. J. 359 (2019) 363–372, https://doi.org/10.1016/j.cej.2018.11.087.
[5] S.W. Lv, J.M. Liu, C.Y. Li, N. Zhao, Z.H. Wang, S. Wang, A novel and universal

metal-organic frameworks sensing platform for selective detection and efficient removal of heavy metal ions, Chem. Eng. J. 375 (2019) 122111, , https://doi.org/10.1016/j.cej.2019.122111.
[6] T. Phothitontimongkol, K. Sanuwong, N. Siebers, N. Sukpirom, F. Unob, Functionalized hectorite clay mineral for Ag(I) ions extraction from wastewater and preparation of silver nanoparticles supported clay, Appl. Clay Sci. 80-81 (2013) 346–350, https://doi.org/10.1016/j.clay.2013.06.002.
[7] F.G.L.M. Borsagli, A.A.P. Mansur, P. Chagas, L.C.A. Oliveira, H.S. Mansur, O-carboxymethyl functionalization of chitosan: complexation and adsorption of Cd (II) and Cr (VI) as heavy metal pollutant ions, React. Funct. Polym. 97 (2015) 37–47, https://doi.org/10.1016/j.reactfunctpolym.2015.10.005.
[8] X. Hu, Y. Hu, G. Xu, M. Li, Y. Zhu, L. Jiang, Y. Tu, X. Zhu, X. Xie, A. Li, Green synthesis of a magnetic β-cyclodextrin polymer for rapid removal of organic micropollutants and heavy metals from dyeing wastewater, Environ. Res. (2019) 108796, , https://doi.org/10.1016/j.envres.2019.108796.
[9] M.R. Awual, M.M. Hasan, G.E. Eldesoky, M.A. Khaleque, M.M. Rahman, M. Naushad, Facile mercury detection and removal from aqueous media involving ligand impregnated conjugate nanomaterials, Chem. Eng. J. 290 (2016) 243–251, https://doi.org/10.1016/j.cej.2016.01.038.
[10] M.R. Awual, M. Ismael, T. Yaita, S.A. El-Safty, H. Shiwaku, Y. Okamoto, S. Suzuki, Trace copper(II) ions detection and removal from water using novel ligand modified composite adsorbent, Chem. Eng. J. 222 (2013) 67–76, https://doi.org/10.1016/j.cej.2013.02.042.
[11] A. Corami, S. Mignardi, V. Ferrini, Cadmium removal from single- and multi-metal (Cd + Pb + Zn + Cu) solutions by sorption on hydroxyapatite, J. Colloid Interf. Sci. 317 (2008) 402–408, https://doi.org/10.1016/j.jcis.2007.09.075.
[12] B. Sandrine, N. Ange, B.A. Didier, C. Eric, S. Patrick, Removal of aqueous lead ions by hydroxyapatites: equilibria and kinetic processes, J. Hazard. Mater. 139 (2007) 443–446, https://doi.org/10.1016/j.jhazmat.2006.02.039.
[13] Y. Feng, J.L. Gong, G.M. Zeng, Q.Y. Niu, H.-Y. Zhang, C.-G. Niu, J.H. Deng, M. Yan, Adsorption of Cd (II) and Zn (II) from aqueous solutions using magnetic hydroxyapatite nanoparticles as adsorbents, Chem. Eng. J. 162 (2010) 487–494, https://doi.org/10.1016/j.cej.2010.05.049.
[14] R. Zhu, R. Yu, J. Yao, D. Mao, C. Xing, D. Wang, Removal of Cd^{2+} from aqueous solutions by hydroxyapatite, Catal. Today 139 (2008) 94–99, https://doi.org/10.1016/j.cattod.2008.08.011.
[15] M. Ozawa, S. Kanahara, Removal of aqueous lead by fish-bone waste hydroxyapatite powder, J. Mater. Sci. 40 (2005) 1037–1038, https://doi.org/10.1007/s10853-005-6529-9.
[16] Y.C. Qi, J. Shen, Q.Y. Jiang, B. Jin, J.W. Chen, X. Zhang, J.L. Su, Hierarchical porous hydroxyapatite microspheres: synthesis and application in water treatment, J. Mater. Sci. 51 (2016) 2598–2607, https://doi.org/10.1007/s10853-015-9573-0.
[17] C. Stötzel, F.A. Müller, F. Reinert, F. Niederdraenk, J.E. Barralet, U. Gbureck, Ion adsorption behaviour of hydroxyapatite with different crystallinities, Colloids Surf. B Biointerfaces 74 (2009) 91–95, https://doi.org/10.1016/j.colsurfb.2009.06.031.
[18] G.J. Ding, Y.J. Zhu, C. Qi, T.W. Sun, J. Wu, F. Chen, Amorphous calcium phosphate nanowires prepared using beta-glycerophosphate disodium salt as an organic phosphate source by a microwave-assisted hydrothermal method and adsorption of heavy metals in water treatment, RSC Adv. 5 (2015), https://doi.org/10.1039/c5ra04624f.
[19] C. Combes, C. Rey, Amorphous calcium phosphates: synthesis, properties and uses in biomaterials, Acta Biomater. 6 (2010) 3362–3378, https://doi.org/10.1016/j.actbio.2010.02.017.
[20] M. Maciejewski, T.J. Brunner, S.F. Loher, W.J. Stark, A. Baiker, Phase transitions in amorphous calcium phosphates with different Ca/P ratios, Thermochim. Acta 468 (2008) 75–80, https://doi.org/10.1016/j.tca.2007.11.022.
[21] S. Somrani, M. Banu, M. Jemal, C. Rey, Physico-chemical and thermochemical studies of the hydrolytic conversion of amorphous tricalcium phosphate into apatite, J. Solid State Chem. 178 (2005) 1337–1348, https://doi.org/10.1016/j.jssc.2004.11.029.
[22] G. Williams, J.D. Sallis, Structure-activity relationship of inhibitors of hydroxyapatite formation, Biochem. J. 184 (1979) 181–184, https://doi.org/10.1042/bj1840181.
[23] L.W. Du, S. Bian, B.D. Gou, Y. Jiang, J. Huang, Y.X. Gao, Y.D. Zhao, W. Wen, T.L. Zhang, K. Wang, Structure of clusters and formation of amorphous calcium phosphate and hydroxyapatite: from the perspective of coordination chemistry, Cryst. Growth Des. 13 (2013) 3103–3109, https://doi.org/10.1021/cg400498j.
[24] M. Karimi, S. Hesaraki, M. Alizadeh, A. Kazemzadeh, A facile and sustainable method based on deep eutectic solvents toward synthesis of amorphous calcium phosphate nanoparticles: the effect of using various solvents and precursors on physical characteristics, J. Non-Cryst. Solids. 443 (2016) 59–64, https://doi.org/10.1016/j.jnoncrysol.2016.04.026.
[25] G.J. Ding, Y.J. Zhu, C. Qi, B.Q. Lu, J. Wu, F. Chen, Porous microspheres of amorphous calcium phosphate: block copolymer templated microwave-assisted hydrothermal synthesis and application in drug delivery, J. Colloid Interf. Sci. 443 (2015) 72–79, https://doi.org/10.1016/j.jcis.2014.12.004.
[26] O.M. Clarkin, M.R. Towler, G.M. Insley, M.E. Murphy, Phase transformations of calcium phosphates formed in wet field environments, J. Mater. Sci. 42 (2007) 8357–8362, https://doi.org/10.1007/s10853-006-0783-3.
[27] F. Zhuang, R. Tan, W. Shen, X. Zhang, W. Xu, W. Song, Monodisperse magnetic hydroxyapatite/Fe_3O_4 microspheres for removal of lead(II) from aqueous solution, J. Alloys. Compd. 637 (2015) 531–537, https://doi.org/10.1016/j.jallcom.2015.02.216.
[28] H. Yang, S. Masse, H. Zhang, C. Hélary, L. Li, T. Coradin, Surface reactivity of hydroxyapatite nanocoatings deposited on iron oxide magnetic spheres toward toxic metals, J. Colloid Interf. Sci. 417 (2014) 1–8, https://doi.org/10.1016/j.jcis.2013.11.031.
[29] S. Rout, B. Behera, T. Maiti, S. Mohapatra, Multifunctional magnetic calcium phosphate nanoparticles for targeted platin delivery, Dalton Trans. 41 (2012) 10777–10783, https://doi.org/10.1039/c2dt30984j.
[30] B.Q. Lu, Y.J. Zhu, F. Chen, C. Qi, X. Zhao, J. Zhao, Core-Shell hollow microspheres of magnetic iron oxide@amorphous calcium phosphate: synthesis using adenosine 5' -triphosphate and application in pH-responsive drug delivery, Chem. Asian J. 9 (2014), https://doi.org/10.1002/asia.201402319.
[31] S. Saoiabi, K. Achelhi, S. Masse, A. Saoiabi, A. Laghzizil, T. Coradin, Organo-apatites for lead removal from aqueous solutions: a comparison between carboxylic acid and aminophosphonate surface modification, Colloids Surface A. 419 (2013) 180–185, https://doi.org/10.1016/j.colsurfa.2012.12.005.
[32] S. Goss, J. Prushko, R. Bogner, Factors affecting calcium precipitation during neutralisation in a simulated intestinal environment, J. Pharm. Sci. 99 (2010) 4183–4191, https://doi.org/10.1002/jps.22131.
[33] Y.Y. Hu, X.P. Liu, X. Ma, A. Rawal, T. Prozorov, M. Akinc, S.K. Mallapragada, K. Schmidt-Rohr, Biomimetic self-assembling copolymer – hydroxyapatite nanocomposites with the nanocrystal size controlled by citrate, Chem. Mater. 23 (2011) 2481–2490, https://doi.org/10.1021/cm200355n.
[34] K. Achelhi, S. Masse, G. Laurent, A. Saoiabi, A. Laghzizil, T. Coradin, Role of carboxylate chelating agents on the chemical, structural and textural properties of hydroxyapatite, Dalton Trans. 39 (2010) 10644–10651, https://doi.org/10.1039/c0dt00251h.
[35] M.E. de Sousa, M.B. Fernández van Raap, P.C. Rivas, P. Mendoza Zélis, P. Girardin, G.A. Pasquevich, J.L. Alessandrini, D. Muraca, F.H. Sánchez, Stability and relaxation mechanisms of citric acid coated magnetite nanoparticles for magnetic hyperthermia, J. Phys. Chem. C. 117 (2013) 5436–5445, https://doi.org/10.1021/jp311556b.
[36] S. Suvokhiaw, A. Imyim, N. Sukpirom, As(V) removal using a magnetic layered double hydroxide composite, Sep. Sci. Technol. 51 (2016) 2948–2957, https://doi.org/10.1080/01496395.2016.1231206.
[37] P. Jantaratana, J. Noodam, C. Sirisathitkul, Magnetic hysteresis and electrical impedance spectra of hard magnetic $SmCo_5$ and soft magnetic $Co_{30}Ag_{70}$ composites, Rare Metal Mat. Eng. 42 (2013) 19–22, https://doi.org/10.1016/S1875-5372(13)60030-2.
[38] S. Recillas, V. Rodriguez Lugo, M. Montero, S. Viquez, L. Hernandez, V. Castaño, Studies on the precipitation behavior of calcium phosphate solutions, J. Ceram. Process. Res. 13 (2012) 5–10.
[39] Z. Stachurski, On structure and properties of amorphous materials, Materials 4 (2011) 1564–1598, https://doi.org/10.3390/ma4091564.
[40] Y. Cai, P. Storer, A. Kheifets, I.E. McCarthy, E. Weigold, Dangling-bond surface states on an amorphous germanium surface as observed by (e,2e) spectroscopy, Surf. Sci. 358 (1996) 427–431, https://doi.org/10.1016/0039-6028(96)00193-8.
[41] V. Patsula, L. Kosinová, M. Lovrić, L. Ferhatovic Hamzić, M. Rabyk, R. Konefal, A. Paruzel, M. Šlouf, V. Herynek, S. Gajović, D. Horák, Superparamagnetic Fe_3O_4 nanoparticles: synthesis by thermal decomposition of iron(III) glucuronate and application in magnetic resonance imaging, ACS Appl. Mater. Interfaces 8 (2016) 7238–7247, https://doi.org/10.1021/acsami.5b12720.
[42] R.Y. Hong, J.H. Li, S.Z. Zhang, H.Z. Li, Y. Zheng, J.M. Ding, D.G. Wei, Preparation and characterization of silica-coated Fe_3O_4 nanoparticles used as precursor of ferrofluids, Appl. Surf. Sci. 255 (2009) 3485–3492, https://doi.org/10.1016/j.apsusc.2008.09.071.
[43] J. Pan, L. Xu, J. Dai, X. Li, H. Hang, P. Huo, C. Li, Y. Yan, Magnetic molecularly imprinted polymers based on attapulgite/Fe_3O_4 particles for the selective recognition of 2,4-dichlorophenol, Chem. Eng. J. 174 (2011) 68–75, https://doi.org/10.1016/j.cej.2011.08.046.
[44] S.K. Milonjić, M.M. Kopečni, Z.E. Ilić, The point of zero charge and adsorption properties of natural magnetite, J. Radioanal. Chem. 78 (1983) 15–24, https://doi.org/10.1007/bf02519745.
[45] S. Rajput, C.U. Pittman, D. Mohan, Magnetic magnetite (Fe_3O_4) nanoparticle synthesis and applications for lead (Pb^{2+}) and chromium (Cr^{6+}) removal from water, J. Colloid Interf. Sci. 468 (2016) 334–346, https://doi.org/10.1016/j.jcis.2015.12.008.
[46] D. Wyrzykowski, E. Hebanowska, G. Nowak-Wiczk, M. Makowski, L. Chmurzyński, Thermal behaviour of citric acid and isomeric aconitic acids, J. Therm. Anal. Calorim. 104 (2011) 731–735, https://doi.org/10.1007/s10973-010-1015-2.
[47] O.A. Pinto, A. Tabaković, T.M. Goff, Y. Liu, J.H. Adair, Calcium phosphate and calcium phosphosilicate mediated drug delivery and imaging, Intracellular Delivery: Fundamentals and Applications (2011) 713–744, https://doi.org/10.1007/978-94-007-1248-5_23.
[48] C. LC, E. Eanes, O. Phosplrate, Solubility of calcium phosphates, Monogr. Oral Sci. 18 (2001) 94–111.
[49] S. Sharma, C. Sebwato, B. Petrusevski, J. Schippers, Effect of groundwater quality on adsorptive iron removal, J. Water Supply Res. T. 51 (2002) 199–208, https://doi.org/10.2166/aqua.2002.0017.
[50] M.R. Awual, T. Yaita, T. Taguchi, H. Shiwaku, S. Suzuki, Y. Okamoto, Selective cesium removal from radioactive liquid waste by crown ether immobilized new class conjugate adsorbent, J. Hazard. Mater. 278 (2014) 227–235, https://doi.org/10.1016/j.jhazmat.2014.06.011.
[51] I. Smičiklas, A. Onjia, S. Raičević, Đ. Janaćković, M. Mitrić, Factors influencing the removal of divalent cations by hydroxyapatite, J. Hazard. Mater. 152 (2008) 876–884, https://doi.org/10.1016/j.jhazmat.2007.07.056.
[52] M. Fehr, A. Schnegg, B. Rech, K. Lips, O. Astakhov, F. Finger, C. Freysoldt, R. Bittl, C. Teutloff, Dangling bonds in amorphous silicon investigated by multifrequency EPR, J. Non-Cryst. Solids. 358 (2012) 2067–2070, https://doi.org/10.1016/j.jnoncrysol.2011.12.105.

[53] C. Freysoldt, G. Pfanner, J. Neugebauer, The dangling-bond defect in amorphous silicon: statistical random versus kinetically driven defect geometries, J. Non-Cryst. Solids. 358 (2012) 2063–2066, https://doi.org/10.1016/j.jnoncrysol.2011.12.090.
[54] M.R. Awual, A facile composite material for enhanced cadmium(II) ion capturing from wastewater, J. Environ. Chem. Eng. 7 (2019) 103378, , https://doi.org/10.1016/j.jece.2019.103378.
[55] O. Karnitz, L.V.A. Gurgel, J.C.P. de Melo, V.R. Botaro, T.M.S. Melo, R.P. de Freitas Gil, L.F. Gil, Adsorption of heavy metal ion from aqueous single metal solution by chemically modified sugarcane bagasse, Bioresource Technol. 98 (2007) 1291–1297, https://doi.org/10.1016/j.biortech.2006.05.013.
[56] H. Benaïssa, Screening of new sorbent materials for cadmium removal from aqueous solutions, J. Hazard. Mater. 132 (2006) 189–195, https://doi.org/10.1016/j.jhazmat.2005.07.085.